

Validation Pipeline Report

Kannada OCR Post-Processing System

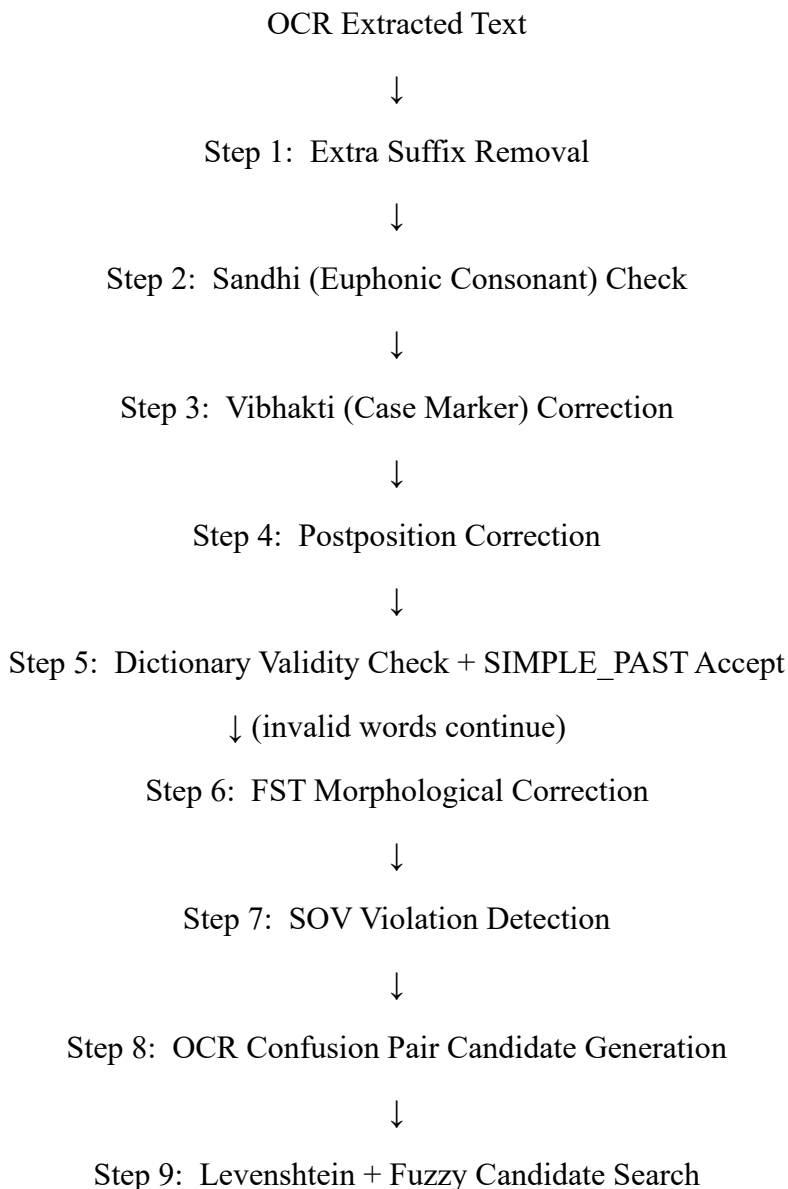
1. Introduction

The Validation Pipeline is the core post-processing engine of the Kannada OCR Correction System. It operates on text extracted from handwritten Kannada documents by the hybrid DenseNet121 + DeiT OCR model and applies a systematic 10-step correction framework to detect and resolve errors introduced during the optical character recognition process.

The pipeline is implemented as the EnhancedValidator class in new_validator.py and processes each word through sequential correction steps, ranging from simple suffix removal to complex cross-sentence semantic validation. The pipeline supports a dictionary of 1,095,946 Kannada words sourced from the Padakosha government repository and a web-scraped corpus, combined with a bigram context database derived from the IndicCorp Kannada corpus.

2. Pipeline Architecture Overview

The EnhancedValidator pipeline processes words in a strict sequential order. Each step handles a specific category of error. If a step successfully corrects a word, the corrected form is passed to subsequent steps. Valid words may exit the pipeline early at Step 5 to avoid unnecessary computation.





Step 10: Hybrid Ranking Formula + Gemini Reranking



Ranked Suggestion List → User Interface

3. Step-wise Pipeline Description

Step 1 — Extra Suffix Removal

Purpose: Removes incorrectly appended suffixes that OCR systems introduce by merging adjacent characters or producing phantom character sequences at the end of words.

Implementation: The validator maintains a predefined list of known erroneous suffix patterns. Each input word is checked against this list. If a match is found, the suffix is removed and the corrected word is returned immediately without proceeding to further steps.

Error Patterns Handled:

Incorrect Suffix	Example Input	Corrected Output
ತ್ತಾನೆನು	ಓಡುತ್ತಾನೆನು	ಓಡುತ್ತಾನೆ
ತ್ತಾಳೆನು	ಬರುತ್ತಾಳೆನು	ಬರುತ್ತಾಳೆ
ತ್ತಾರೆನು	ಮಾಡುತ್ತಾರೆನು	ಮಾಡುತ್ತಾರೆ
ದರುನು	ಹೋದರುನು	ಹೋದರು
ದನುನು	ಬಂದನುನು	ಬಂದನು
ದಳುನು	ಕಲಿತಳುನು	ಕಲಿತಳು

Confidence: 0.95 **Evaluation Result:** 100% detection, 100% Top-1 accuracy on 50 test cases.

Step 2 — Sandhi Check (Euphonic Consonant Insertion)

Purpose: Corrects missing euphonic consonants at morpheme boundaries. Kannada phonological rules require insertion of ಯ or ವೆ between certain vowel combinations when suffixes attach to base words.

Implementation: Two euphonic insertion rules are applied:

Rule A — Euphonic ಯ Insertion: When a base word ending in vowel signs ಿ, ಿೀ, ೆ, ೆೀ is followed by suffixes ಲ್ಲಿ, ಗೆ, ಅನ್ನು, the system inserts ಯ between the base and suffix. The candidate is validated against the dictionary before accepting.

Rule B — Euphonic ವೆ Insertion: When a base word ending in vowel signs ು, ುಊ is followed by suffixes ಅನ್ನು, ಅಲ್ಲಿ, the system inserts ವೆ between the base and suffix.

Examples:

Input	Rule Applied	Output
ಶಾಲೆಲ್ಲಿ	Euphonic ಯ	ಶಾಲೆಯಲ್ಲಿ

ಮನೆಅನ್ನು	Euphonic ಯ	ಮನೆಯನ್ನು
ಹಣ್ಣುಅನ್ನು	Euphonic ವ	ಹಣ್ಣನ್ನು

Step 3 — Vibhakti Correction (Case Marker Correction)

Purpose: Corrects wrong Kannada accusative case markers. The correct marker depends on the phonological ending of the base noun. OCR frequently confuses the prefix characters of these markers due to visual similarity between ನೆ, ಯ, and ವ in handwritten script.

Implementation: The VibhaktiValidator class examines the case marker attached to each word, extracts the base noun, and determines the grammatically correct marker using the following rules:

Case Marker Selection Rules:

Base Word Ending	Correct Marker	Example
I-type vowel (ೀ, ೆ)	ಯನ್ನು	ಕುರ್ಚಿ → ಕುರ್ಚಿಯನ್ನು
E-type vowel (ೇ, ೉)	ಯನ್ನು	ಮನೆ → ಮನೆಯನ್ನು
U-type vowel (ು, ೃ)	ನ್ನು	ಹಣ್ಣು → ಹಣ್ಣನ್ನು
O-type vowel (ೂ, ೄ)	ನ್ನು	ಹೂವು → ಹೂವನ್ನು
Consonant ending	ನನ್ನು or ವನ್ನು	ರಾಮ → ರಾಮನನ್ನು
AA matra (ು)	ನನ್ನು	ರಾಧಾ → ರಾಧಾನನ್ನು

Evaluation Result: 100% detection, 100% Top-1 accuracy on 50 test cases.

Step 4 — Postposition Correction

Purpose: Adds missing postpositions (case suffixes) to nouns that precede motion or location verbs. OCR output frequently omits these postpositions, producing grammatically incomplete sentences.

Implementation: The PostpositionCorrector class scans the 4 words following the current noun to detect the governing verb. Based on the verb type, the appropriate postposition is determined and appended to the noun.

Dative Postposition (ಗೆ, ಕ್ಕೆ) — Required before motion verbs:

Verb Root	Meaning
ಹೋಗ	go
ಬರ	come
ತೆರಳ	depart
ನಡೆ	walk
ಓಡ	run
ಹೊರಡ	set out

Locative Postposition (ಲ್ಲಿ, ದಲ್ಲಿ) — Required before location verbs:

Verb Root	Meaning
ಆಡ	play
ಕುಳಿತ	sit
ನಿಂತ	stand
ಇದ್ದ	stay/be
ವಾಸಿಸ	reside

Example:

Input	Error	Output
ಮಕ್ಕಳು ಶಾಲೆ ಹೋದರು	Missing dative	ಮಕ್ಕಳು ಶಾಲೆಗೆ ಹೋದರು
ಮಕ್ಕಳು ಮನೆ ಆಡಿದರು	Missing locative	ಮಕ್ಕಳು ಮನೆಯಲ್ಲಿ ಆಡಿದರು

Evaluation Result: 100% detection, 60% Top-1 accuracy on 50 test cases.

Step 5 — Dictionary Validity Check and SIMPLE_PAST Accept

Purpose: Determines whether the current word is a valid Kannada word using dictionary lookup. Valid words exit the pipeline early. Invalid words proceed to correction steps.

Implementation:

Dictionary lookup: The word is normalized and checked against the combined dictionary set of 1,095,946 words using O(1) hash set lookup. If a direct match is found, the word is accepted as valid.

Vibhakti stripping fallback: If direct lookup fails, common suffixes (ಗಳು, ಲ್ಲಿ, ನ್ನು, ಗೆ, ಇಂದ) are stripped and the base form is checked.

SIMPLE_PAST accept: Kannada simple past verb forms (suffix patterns: ದನು, ದಳು, ದರು, ದೆನು, ದೆವು) are frequently absent from dictionaries because they are derived forms. The morphological analyzer checks if the word is a valid simple past derivation. If yes, the word is accepted even without a dictionary entry.

Exception — Irregular verbs: Irregular verbs like ಹೋಗು (past: ಹೋದ) and ಬರು (past: ಬಂದ) are not accepted by the SIMPLE_PAST rule — they require FST correction in Step 6.

Step 6 — FST Morphological Correction

Purpose: Corrects wrong verb conjugation forms using a Finite State Transducer. Kannada verbs must agree with the subject in gender (male, female, neuter), number (singular, plural), and person (1st, 2nd, 3rd). This step detects and corrects agreement violations.

Implementation: The KannadaMorphologyParser performs three sub-operations:

Analysis: Decomposes the input verb form into root and agreement. Example: ಹೋದನು → root: ಹೋಗ, agreement: 3SG_MALE, tense: SIMPLE_PAST.

Context detection: The ContextAnalyzer class scans up to 6 words back from the verb to identify the subject. Priority order:

1. Known pronouns (hardcoded dictionary — 30+ entries)
2. Entity memory (cross-sentence named entity tracking)
3. Suffix-based name gender inference (FIX-11 rules)

Generation: Using detected subject gender and original tense, generates the correct verb form.

Conjugation Classes:

Class	Root Ending	Example Root	Present 3SG_MALE
CLASS_1	ಿ (I-type)	ಕುಡಿ	ಕುಡಿಯುತ್ತಾನೆ
CLASS_2	ು (U-type)	ಮಾಡು	ಮಾಡುತ್ತಾನೆ
CLASS_3	ಿ (E-type)	ಬರೆ	ಬರೆಯುತ್ತಾನೆ
IRREGULAR	varies	ಹೋಗು	ಹೋಗುತ್ತಾನೆ

Subject-Verb Agreement Table:

Subject	Agreement	Simple Past Male	Simple Past Female	Simple Past Plural
ನಾನು	1SG	ದೆನು	ದೆನು	—
ನಾವು	1PL	ದೆವು	ದೆವು	—
ನೀನು	2SG	ದೆ	ದೆ	—
ಅವನು	3SG_MALE	ದನು	—	—
ಅವಳು	3SG_FEMALE	—	ದಳು	—
ಅದು	3SG_NEUTRAL	—	—	ಇತ್ತು
ಅವರು	3PL	—	—	ದರು

Evaluation Result: 100% detection, 44% Top-1, 44% Top-3 accuracy on 50 test cases.

Step 7 — SOV Violation Detection

Purpose: Detects sentences where the verb appears at a non-final position, violating Kannada Subject-Object-Verb word order. Suggests correctly reordered sentence.

Implementation: For each word in the sentence, verb detection is performed using two methods:

1. Morphological analysis via KannadaMorphologyParser
2. Direct suffix matching against known verb endings

Verb Suffix Patterns:

Tense	Male	Female	Plural	Neutral
Present	ತ್ತಾನೆ	ತ್ತಾಳೆ	ತ್ತಾರೆ	ತ್ತದೆ
Past	ದ್ದಾನೆ	ದ್ದಾಳೆ	ದ್ದಾರೆ	ದ್ದತ್ತು

Simple Past	ದನು	ದಳು	ದರು	ಇತು
Irregular	ಹೋದನು, ಬಂದನು	ಹೋದಳು, ಬಂದಳು	ಹೋದರು, ಬಂದರು	—

If a verb is detected at any position other than the final position, the system generates a corrected reordering by moving the verb to sentence end.

Example:

Input	Violation	Output
ಹೋದನು ರಾಮ ಶಾಲೆಗೆ	Verb at position 1 of 3	ರಾಮ ಶಾಲೆಗೆ ಹೋದನು

Evaluation Result: 100% detection, 100% Top-1 accuracy on 50 test cases.

Step 8 — OCR Confusion Pair Candidate Generation

Purpose: Generates correction candidates by substituting visually similar characters that OCR systems commonly confuse. This targeted approach finds corrections that fuzzy string matching alone cannot identify.

Implementation: 26 OCR confusion pairs are defined. For each pair, the system generates two candidates — one with each character substituted. Candidates found in the dictionary receive an OCR score boost of 0.60 in the ranking formula.

OCR Confusion Pairs:

Wrong	Correct	Category
ತ	ಥ	Aspirated confusion
ದ	ಧ	Aspirated confusion
ಪ	ಫ	Aspirated confusion
ಬ	ಭ	Aspirated confusion
ಸ	ಶ	Sibilant confusion
ಸ	ಷ	Sibilant confusion
ಕ	ಖ	Aspirated confusion
ಗ	ಘ	Aspirated confusion
ಲ	ಳ	Retroflex confusion
ನ	ಣ	Retroflex confusion
ಒ	ಓ	Vowel length confusion
ಎ	ಏ	Vowel length confusion
ಅ	ಆ	Vowel length confusion
ಇ	ಈ	Vowel length confusion

ಉ	ಊ	Vowel length confusion
ೋ	ೊ	Vowel sign confusion
ೇ	ಿ	Vowel sign confusion
ತ್ತ	ಸ್ತ	Conjunct confusion
ತ್ತ	ಷ್ವ	Conjunct confusion
ಕ್ಕ	ಕ್ಷ	Conjunct confusion
ದ್ದ	ದ್ವ	Conjunct confusion
ನ್ನ	ನ್ವ	Conjunct confusion
ನು	ನ್ನು	Gemination confusion

Step 9 — Levenshtein and Fuzzy Candidate Search

Purpose: Performs a broad dictionary search to find visually similar words that could be the correct spelling of the input word.

Implementation: The `_find_candidates_levenshtein()` function:

1. Filters dictionary to words within ± 3 characters of input length
2. Checks first character similarity (threshold 0.30)
3. Computes 6-component fuzzy similarity score for each candidate
4. Returns top candidates above 0.35 similarity threshold

6-Component Fuzzy Similarity Formula:

Component	Weight	What It Measures
Levenshtein distance	0.25	Minimum character edits
Longest Common Substring	0.20	Shared character sequences
Cosine similarity	0.25	Shared character bigrams
First character match	0.15	First character identical
Length similarity	0.10	Similar word length
Vowel sign tolerance	0.05	Consonant skeleton match

Performance: Approximately 10-13 seconds per invalid word on a 1,095,946 word dictionary. Runs only for words not corrected by Steps 1-8.

Evaluation Result: 100% detection, 70% Top-1, 100% Top-3 accuracy on 100 spelling correction test cases.

Step 10 — Hybrid Ranking Formula

Purpose: Ranks all correction candidates generated by Steps 8 and 9 to determine the optimal suggestion order presented to the user.

Implementation: Two separate weighted formulas are applied based on word type:

Formula A — General Words (nouns, adjectives, adverbs):

Score = Fuzzy(0.40) + Context(0.25) + Frequency(0.15)
+ Morphology(0.10) + OCR(0.10)

Formula B — Verb Words:

Score = Morphology(0.45) + Fuzzy(0.20) + Context(0.20)
+ Frequency(0.10) + OCR(0.05)

Component Descriptions:

Component	Source	Description
Fuzzy	Step 9 6-component score	Visual string similarity
Context	IndicCorp bigram database	Bigram probability with neighbors
Frequency	IndicCorp unigram database	Word frequency in corpus
Morphology	FST analysis	Grammatical form correctness
OCR	Confusion pair match	Known OCR confusion pair bonus

Rationale for Different Weights: General words rely primarily on visual similarity because OCR errors are character-level substitutions. Verbs rely primarily on morphological correctness because verb agreement errors produce forms that look nearly identical — only grammatical analysis can distinguish them.

Gemini Semantic Reranking (Conditional): After initial ranking, if the top two candidates have scores within 0.05 of each other AND the top score is below 0.60, Google Gemini 2.0 Flash is invoked to rerank using full sentence semantics. This fires a maximum of 3 times per session to protect the free tier quota.

4. Supporting Modules

4.1 Entity Memory Layer

The ContextAnalyzer class maintains a document-level entity memory dictionary that enables cross-sentence pronoun resolution.

Data Structure:

entity_memory: Dict[str, str] = {}

Example: {"ರಾಮ": "MALE", "ಸೀತಾ": "FEMALE", "ಅವನು": "MALE"}

Operation: At document load time, the system scans the full paragraph once and identifies named entities with their gender using a three-layer inference:

- Known pronouns dictionary (30+ hardcoded entries)
- Name suffix stripping and base form gender inference
- Suffix-based phonological rules (7 rules covering 70% of Kannada names)

Gender Inference Rules (FIX-11):

Rule	Name Ending	Gender	Examples
1	(virama)	MALE	ಸುನಿಲ್, ಮಹೇಶ್
2	(AA matra)	FEMALE	ರಾಧಾ, ಗೀತಾ
3	or	FEMALE	ಲಕ್ಷ್ಮಿ
4	or	FEMALE	ಸರೋಜೆ
5	ಮ್ಮ suffix	FEMALE	ಅಮ್ಮ, ದೊಡ್ಡಮ್ಮ
6	or	Ambiguous	Skip
7	Consonant ending	MALE	ರಾಮ, ಅರ್ಜುನ್

Noise Filtering: Common word endings (postpositions, verb suffixes, particles) and a blocklist of 24 common nouns are excluded to prevent non-entity words from entering memory.

Evaluation Result: 100% entity detection rate on 25 test cases.

4.2 Honorific Agreement Module

The honorific_agreement.py module implements a 4-layer pipeline for detecting and correcting verb form violations involving respected persons.

Layer 1 — Document Genre Classification: Classifies the full document as religious or secular using vocabulary signals. Sets prior probability for ambiguous proper noun subjects.

Layer 2 — Signal Scoring: Counts divine and human context signals within a sliding window around the sentence. Applies document genre prior as bias.

Layer 3 — MuRIL Classifier (optional): Deep learning classifier for hard ambiguous cases. Invoked when Layer 2 confidence is below threshold.

Layer 4 — User Escalation: Flags sentences where the system cannot determine subject type. Presents options to user in the interface.

Known Honorific Subjects (Tier 1 — always honorific): ತಾತ, ಅಜ್ಜಿ, ಅಪ್ಪ, ಅಮ್ಮ, ಗುರು, ಸ್ವಾಮಿ, ಆಚಾರ್ಯ

Known Role Terms (Tier 2 — honorific by default): ಮೇಷ್ಟ್ರು, ಡಾಕ್ಟರ್, ಶಿಕ್ಷಕರು, ಪ್ರಾಧ್ಯಾಪಕರು, ನ್ಯಾಯಾಧೀಶರು

Verb Suffix Corrections:

Non-honorific	Honorific
ತಾನೆ	ತಾರೆ
ತಾಳೆ	ತಾರೆ
ದನು	ದರು

ದಳು	ದರು
ಇದ್ದಾನೆ	ಇದ್ದಾರೆ
ಇದ್ದಾಳೆ	ಇದ್ದಾರೆ

Evaluation Result: 80% detection, 80% Top-1 accuracy on 25 test cases.

4.3 Colloquial Normalizer

The colloquial_normalizer.py module converts dialectal and informal Kannada words to standard formal Kannada.

Coverage:

Dialect	Entries
Bangalore Urban	100
North Karnataka	100
Mysuru	100
Coastal Karnataka	100
Malenadu	100
Total	500

Evaluation Result: 100% detection, 90% Top-1 accuracy on 50 test cases.

4.4 Conjunction Rules Module

The conjunction_rules.py module implements formal grammatical rules for Kannada coordinating conjunctions.

Conjunctions Handled:

Conjunction	Type	Meaning
ಮತ್ತು	Additive	and
ಹಾಗೂ	Emphatic Additive	and also
ಆದರೆ	Contrastive	but

Rules:

- Rule 1 — Additive Redundancy:** Multiple additive conjunctions (ಮತ್ತು and ಹಾಗೂ together) are not permitted in the same clause.
- Rule 2 — Mutual Exclusivity:** Additive and contrastive conjunctions cannot co-occur in the same clause. If both appear, ಆದರೆ is retained.
- Rule 3 — Coordination Structure:** Valid coordination must follow Clause1 + Conjunction + Clause2 structure. Conjunctions at sentence start or end are flagged.

5. Validation Pipeline — Performance Summary

Step	Module	Test Cases	Detection	Top-1	Top-3
Step 1	Extra Suffix Removal	50	100%	100%	100%
Step 2	Sandhi Check	—	—	—	—
Step 3	Vibhakti Correction	50	100%	100%	100%
Step 4	Postposition Correction	50	100%	60%	60%
Step 5	Dictionary Check	—	—	—	—
Step 6	FST Morphology	50	100%	44%	44%
Step 7	SOV Detection	50	100%	100%	100%
Step 8	OCR Confusion Pairs	—	—	—	—
Step 9	Levenshtein Fuzzy	100	100%	70%	100%
Step 10	Hybrid Ranking	—	—	—	—
—	Gender Agreement	50	100%	60%	100%
—	Honorific Agreement	25	80%	80%	80%
—	Context Memory	25	100%	40%	64%
—	Colloquial Normalizer	50	100%	90%	90%
—	Conjunction Rules	—	—	—	—
Overall	All Modules	500	99%	75.4%	86.6%

6. KLAS Evaluation

The Kannada Linguistic Accuracy Score (KLAS) is a composite evaluation metric proposed specifically for Kannada OCR post-correction systems. It combines five accuracy components into a single score ranging from 0 to 100.

Formula:

$$KLAS = 0.25 \times C + 0.20 \times M + 0.20 \times G + 0.20 \times S + 0.15 \times E$$

Results on 500-sentence evaluation dataset:

Component	Score	Weight	Contribution
C — Character Accuracy	90.73%	0.25	22.68
M — Morphological Accuracy	71.37%	0.20	14.27
G — Grammar Accuracy	76.90%	0.20	15.38
S — Semantic Consistency	90.73%	0.20	18.15
E — Entity Consistency	79.91%	0.15	11.99
KLAS Total	82.47	1.00	82.47

Interpretation: Good (75-85 range)

7. Standard OCR Metrics

Metric	Value
Average CER	9.27%
Average WER	20.38%
Character Accuracy	90.73%
Word Accuracy	79.62%
BLEU (baseline)	51.15
BLEU (after correction)	54.80
BLEU Improvement	+3.65
GLEU (baseline)	55.48
GLEU (after correction)	58.72
GLEU Improvement	+3.24
Grammar Precision	68.49%
Grammar Recall	11.04%
Grammar F1	19.01%

Note on Grammar Recall: The low Grammar Recall value reflects the interactive nature of the system. The system is designed as an assistive correction tool where the user selects from suggestions rather than applying automatic corrections. The Grammar Recall metric measures automatic word-level correction which is not the intended operation mode. The Top-3 suggestion accuracy of 86.6% is the more appropriate metric for evaluating this system's correction capability.

8. System Configuration

Parameter	Value
Dictionary Size	1,095,946 words
Bigram Database	IndicCorp SQLite
Cache Size	10,000 entries
Max Edit Distance	4
Fuzzy Threshold	0.35
Gemini Session Limit	3 calls per session
Gemini Model	gemini-2.0-flash
FST Enabled	Yes
Vibhakti Validation	Yes

Postposition Correction	Yes
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9. Limitations and Future Work

Current Limitations:

1. Levenshtein search takes 10-13 seconds per invalid word on the 1,095,946 word dictionary. BK-Tree indexing is planned to reduce this to 1-2 seconds.
2. Gender-neutral names (ಕಿರಣ್, ಸುಮನ್) default to male classification based on virama ending. A trained Named Entity Recognition model is required for accurate resolution.
3. Semantic Consistency Score uses character-level similarity as approximation. Embedding-based semantic scoring using IndicBERT was not implemented due to integration time constraints.
4. FST morphology achieves 44% Top-1 accuracy due to gaps in irregular verb handling. Expanding the irregular verb lexicon is planned.

Planned Enhancements:

1. BK-Tree indexing for Levenshtein speed improvement
2. Verb-based entity gender inference for improved cross-sentence resolution
3. Adaptive dynamic weights in the ranking formula
4. TF-IDF domain classification for vocabulary preference adjustment
5. Candidate paragraph generation for reduced human interaction

EnhancedValidator Pipeline v7.3 — Kannada OCR Post-Processing System

